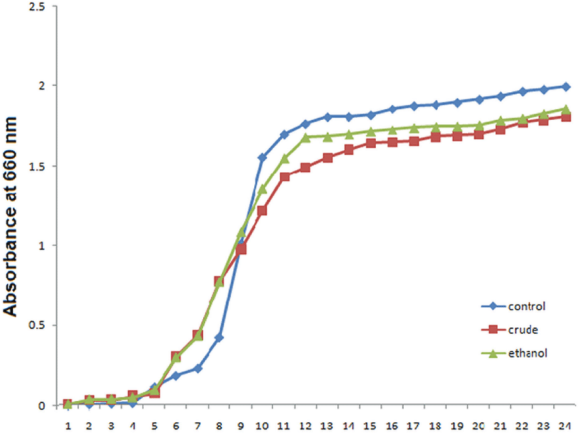




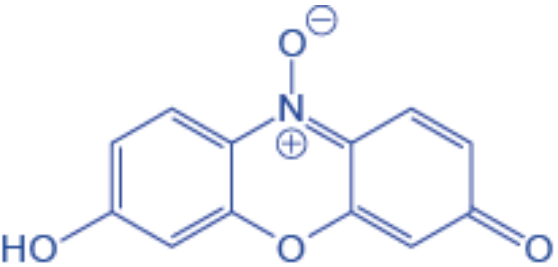
Crash Course on Antimicrobial Peptides in the PechaKucha style





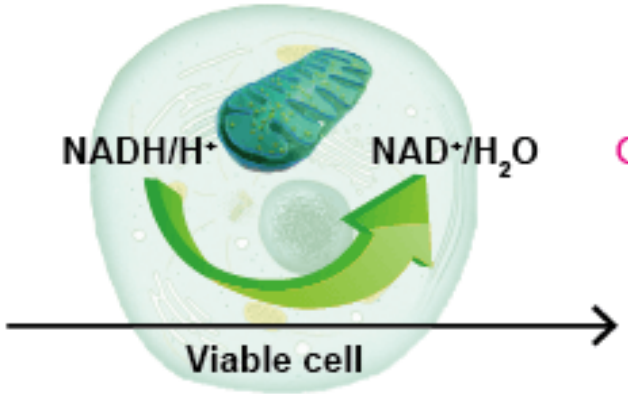
By: Antranig Baghdassarian

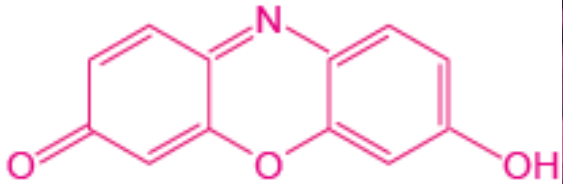
PI: Dr. Candan Tamerler
Mentors: Nur Ceren Kabakci & Aya Kirahm Cloyd



Resazurin

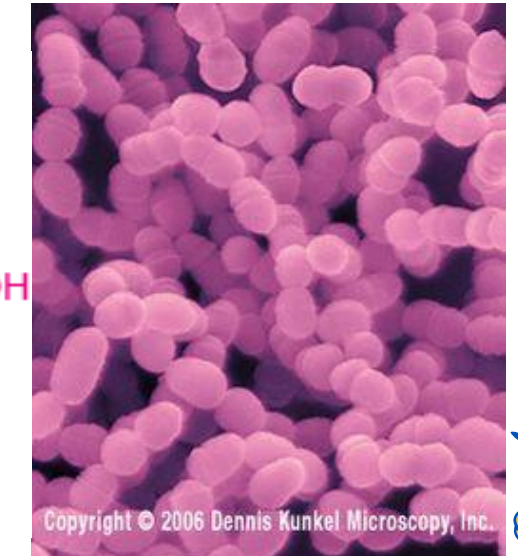
Blue and low fluorescence





Resorufin

Pink and high fluorescence





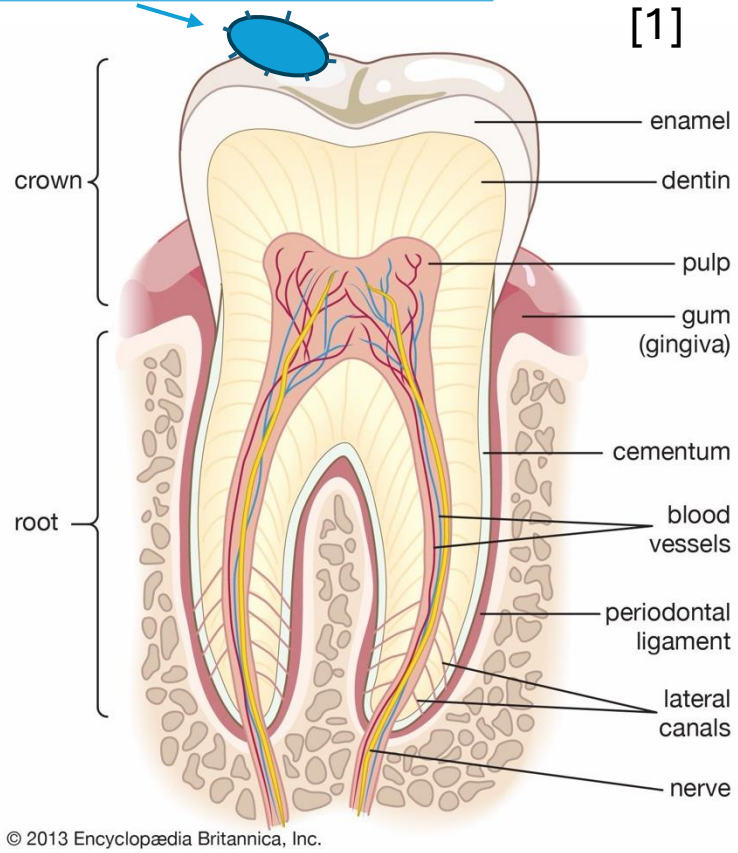
Streptococcus mutans (S.Mutans)
Pictured here! →



A Deeper Look into Cavities…

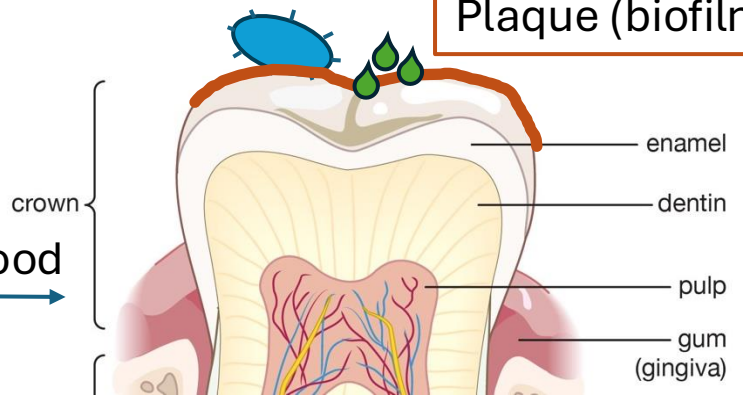
Bacteria (*Streptococcus mutans*) [6,7]

[1]

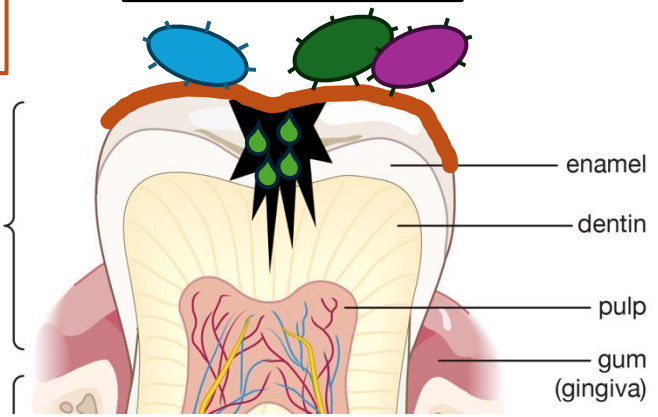


Food

Plaque (biofilm) [3,4,5]



Cavity (caries)



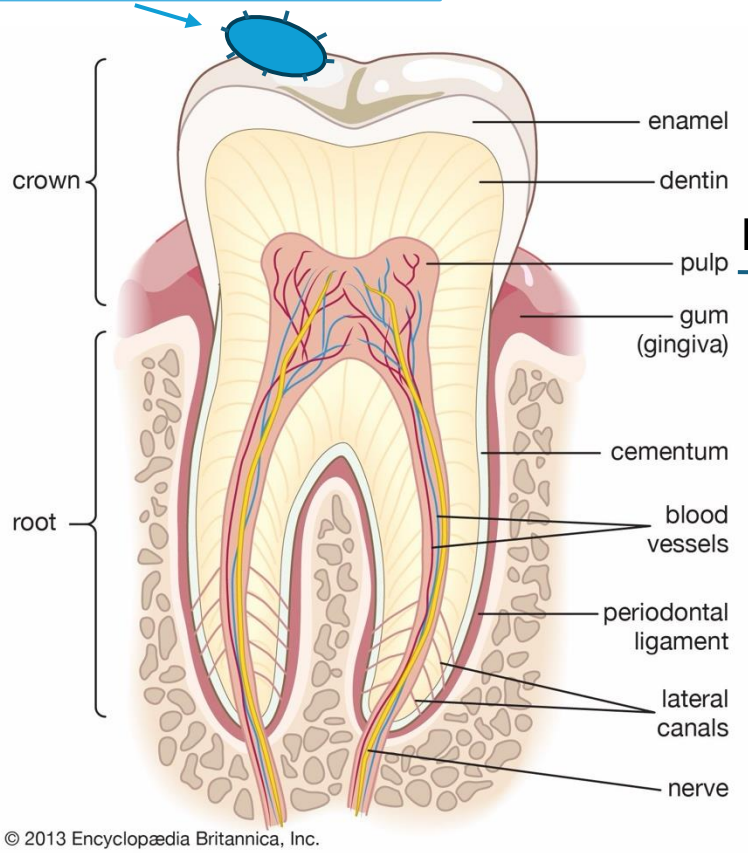
1: https://www.britannica.com/science/tooth-anatomy#/media/1/599469/112882
2: https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/streptococcus-mutans
3: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2292933/#:~:text=mutans%20streptococci%2C%20Actinomyces%20spp.%2C,world%20(2%2C%2039).
4: https://journals.asm.org/doi/10.1128/iai.00422-08
5: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5884221/#:~:text=Streptococcus%20mutans%20has%20been%20implicated,dental%20plaque%20on%20tooth%20surfaces.
6: https://www.ncbi.nlm.nih.gov/books/NBK551699/
7: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6615571/

Streptococcus mutans (S.Mutans)
Pictured here! →



A Deeper Look into Cavities…

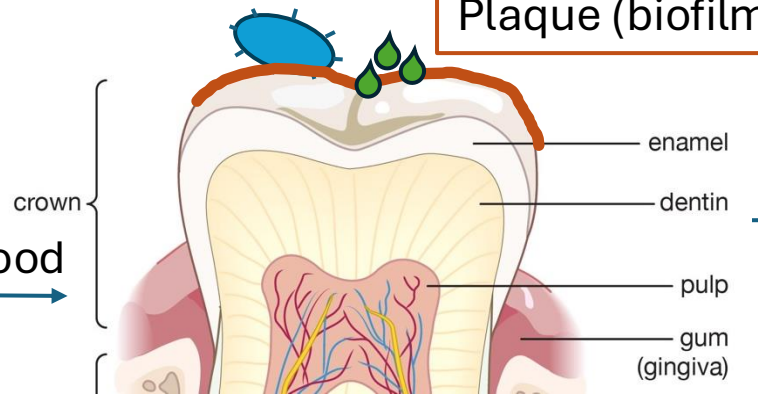
Bacteria (*Streptococcus mutans*)



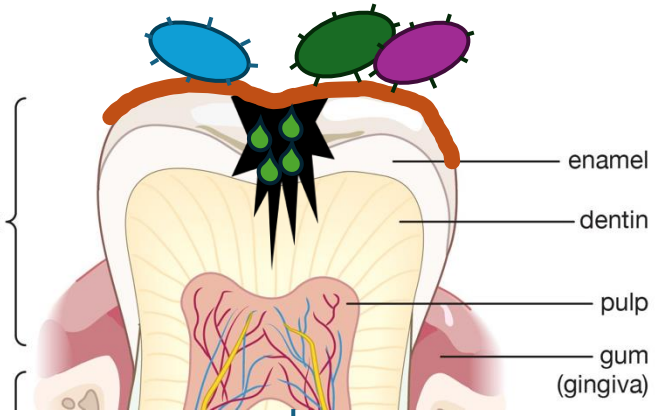
© 2013 Encyclopædia Britannica, Inc.

Food →

Plaque (biofilm)

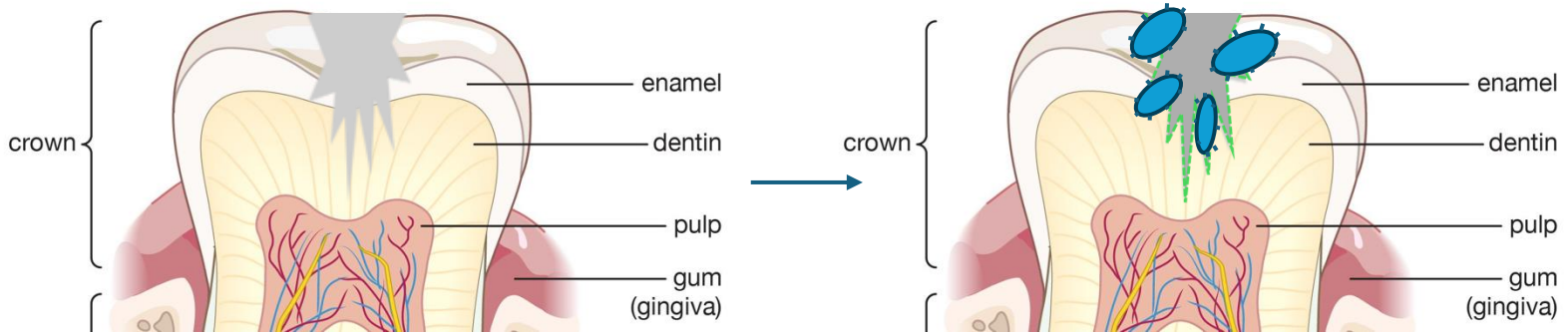


Cavity (caries)



Bacteria in adhesive/dentin interface adhesive 😮

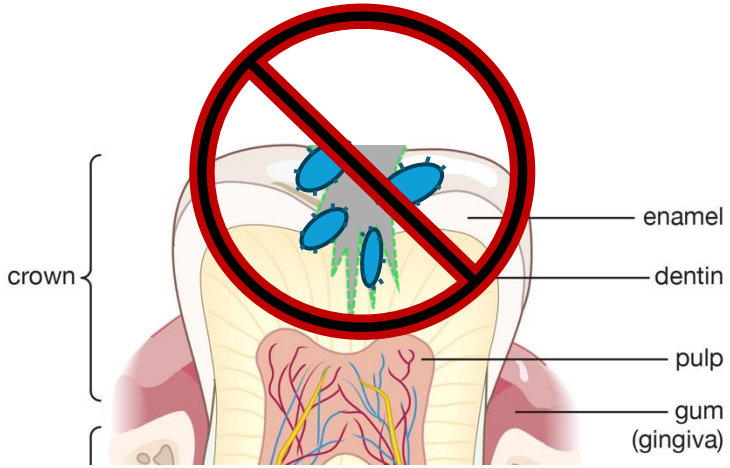
Filling



"bacterial biofilms […] permeate into the adhesive/dentin (a/d) interfacial defects and destroy the adjacent tooth structure" (Sheng-Xue Xie, 2020)

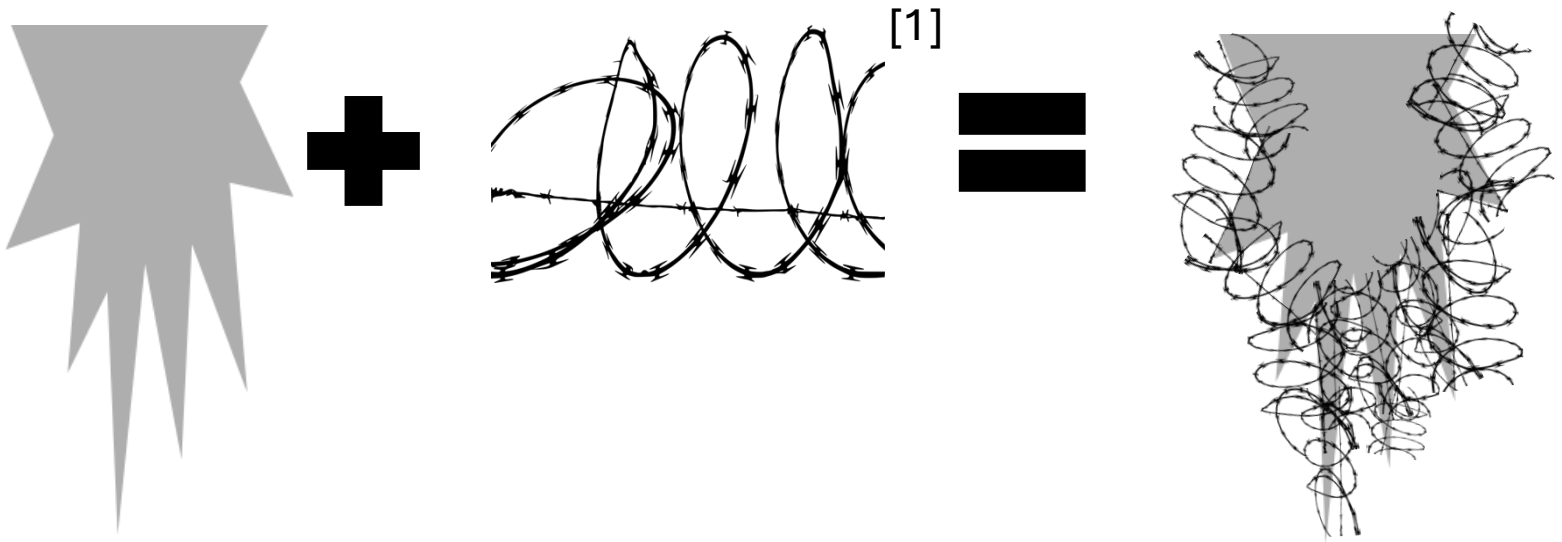
1: https://pubs.acs.org/doi/10.1021/acsapm.9b00921?ref=pdf#[1]

How can we stop this?!?

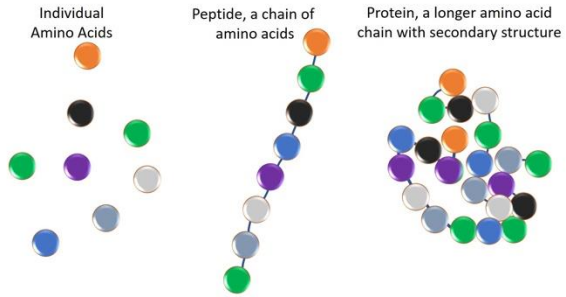


- Antibiotics? ☹
  - Some bacteria are resistant
    - Leads to more bacteria gaining resistance to antibiotics

- What if we build the defense INTO the filling?


[1]

1: https://www.stockio.com/free-clipart/barbwire



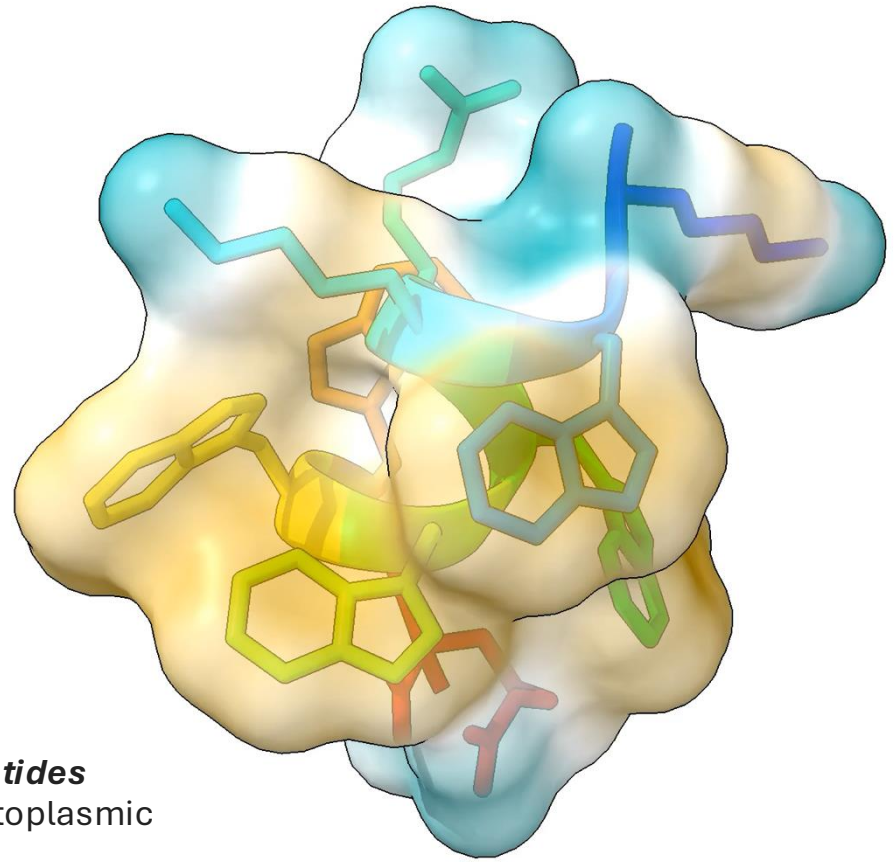
[5]

Introducing… Peptides

- Made of ≤ 50 amino acids
- Peptides can have various functions:
  - **Anti-microbial (AMP)**
  - Anti-fungal
  - Anti-cancer
  - **Collagen binding**
  - Inorganic binding (Calcium, titanium, silver etc.)
- Peptides with ≥ 2 functional groups are <u>chimeric peptides</u>

- Much better than antibiotics!
  - "[I]t is ***very difficult for bacteria to develop resistance to antimicrobial peptides*** because most kill bacterial cells quickly through their actions on the entire cytoplasmic membrane or can act through complex mechanism" (Park S-C, et al. 2011)[2]

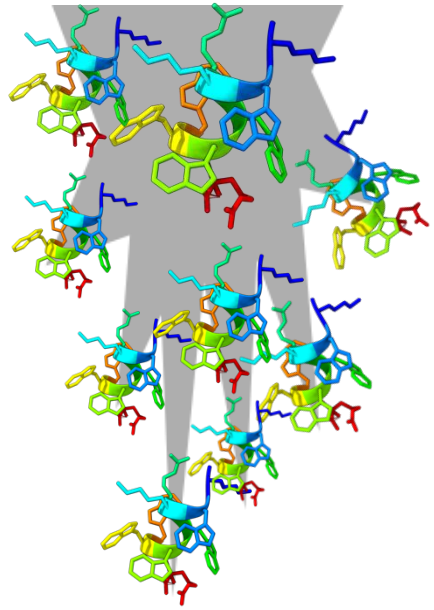
- Antimicrobial + Collagen Binding Peptide:



(Folding simulation of AMP2, used for my experiments!)[1]

1: https://mobyle.rpbs.univ-paris-diderot.fr/cgi-bin/portal.py#welcome
2: https://www.mdpi.com/1422-0067/12/9/5971#:~:text=It%20is%20also%20believed%20that,through%20complex%20mechanism%20%5B12%5D.
3. https://ncbi.nlm.nih.gov/pmc/articles/PMC7609970/
4: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7609970/
5: https://www.cellgs.com/blog/the-difference-between-peptides-and-proteins.html

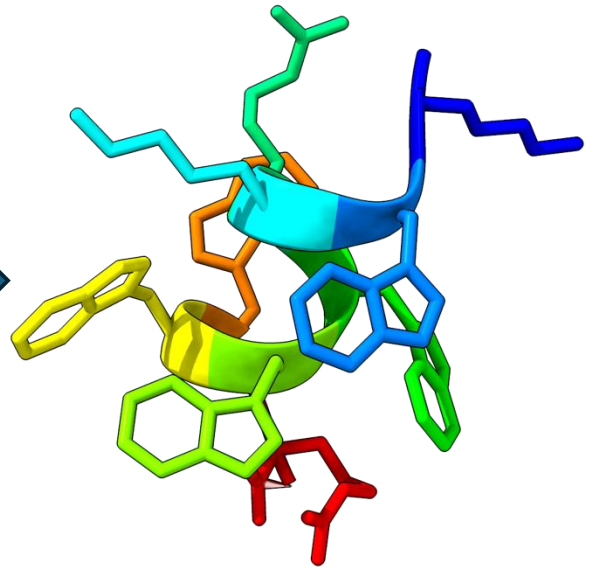
New goal: Dental Filling w/ Peptide:



1. Measure effectiveness of antimicrobial peptide (AMP) against S. Mutans in growth environment

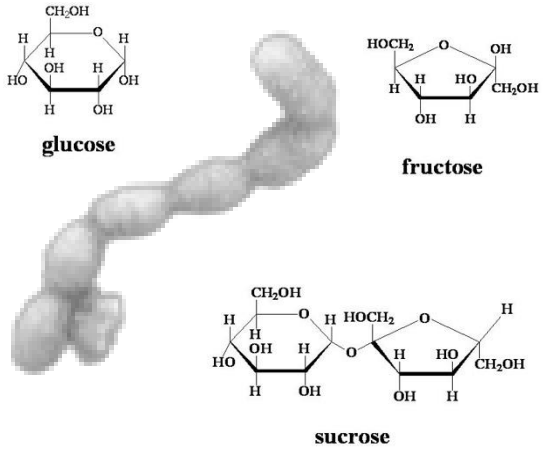


Brain Heart Infusion (BHI)



Antimicrobial + Collagen Binding Peptide





S. Mutans w/ food

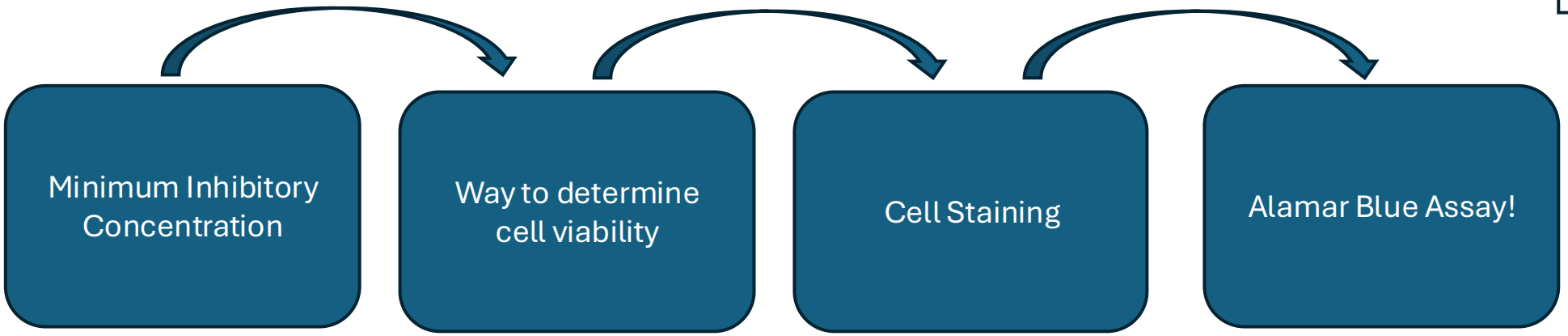
But… How Much Peptides Do We Need?

- Minimum Inhibitory Concentration (MIC): Lowest concentration of AMP where the bacteria stops growing. [1]



[2]

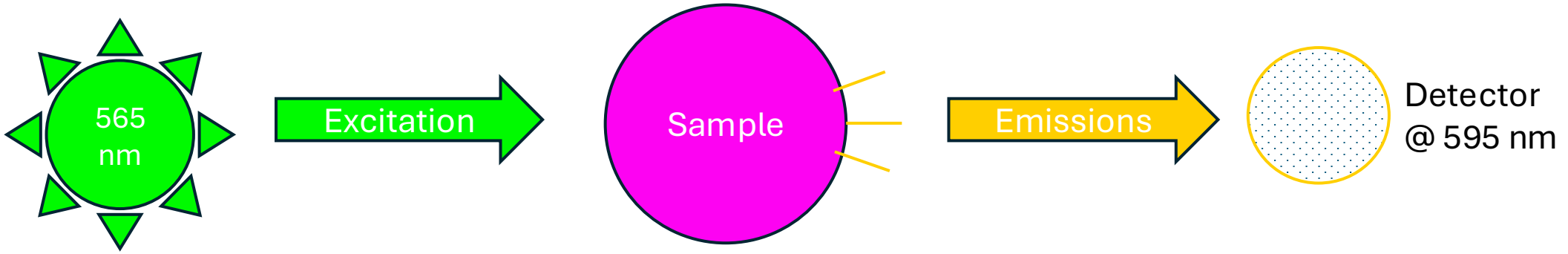
The peptide synthesizer in lab!

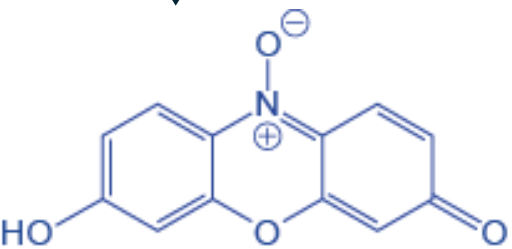


1: https://pubmed.ncbi.nlm.nih.gov/11420333/
2: https://www.peptide.com/peptide-synthesizers/focus-xc/

How Does Alamar Blue Assay Work? [1,2,4,5]

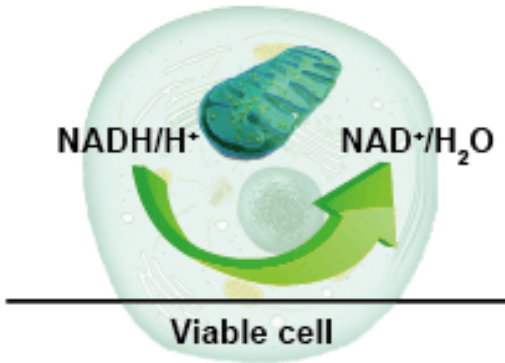
- Measured Fluorescence! The more viable the bacteria, the pinker.



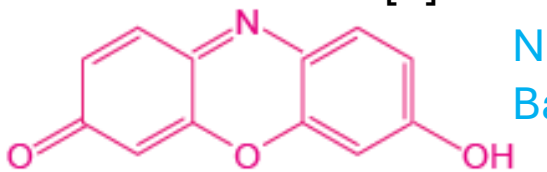


Resazurin

Blue and low fluorescence



[3]



Resorufin

Pink and high fluorescence

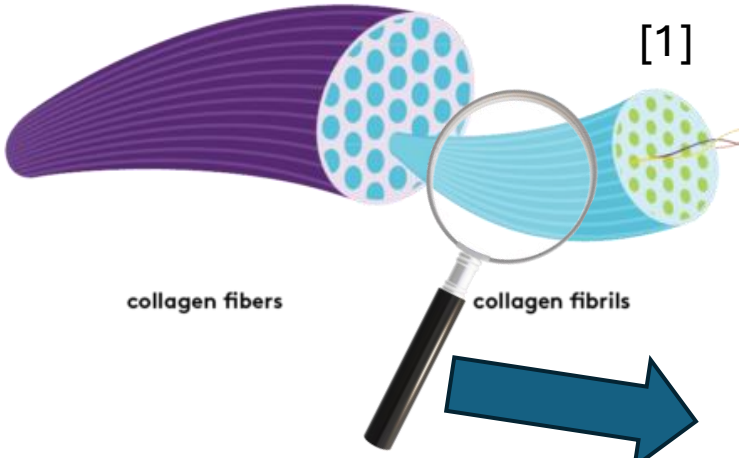


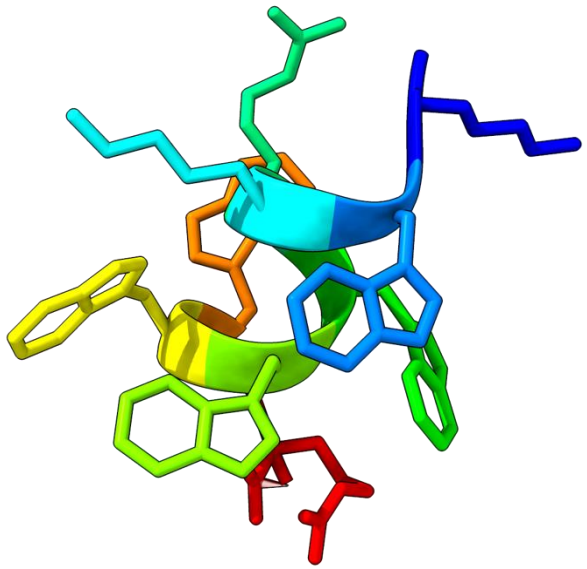
[1] https://www.ncbi.nlm.nih.gov/books/NBK343429/
[2] https://www.sciencedirect.com/topics/medicine-and-dentistry/resazurin-assay
[3] https://www.creative-bioarray.com/support/resazurin-cell-viability-assay.htm
[4] https://www.linkedin.com/pulse/understanding-absorption-emission-excitation-/#:~:text=wavelengths%20of%20light%20that%20a%20fluorophore
[5] https://chem.libretexts.org/Courses/University_of_California_Davis/Chem_4A_Lab%3A_General_Chemistry_for_Majors_I/Chem_4A%3A_Laboratory_Manual/10_7%3A_Quantitative_Spectrophotometry_and_Beer's_Law_(Experiment)

Future Work

- A rematch! (In a different arena)

[1]



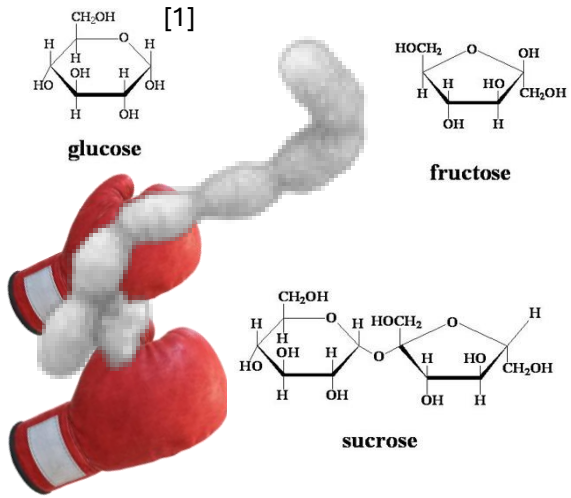


Antimicrobial + **Collagen Binding** Peptide

- Tests if using both functional groups of the peptide changes the efficiency of one or the other

VS

[1]



S. Mutans w/ food

1: https://drkarenwallace.com/blog-drkaren-naturopathic-dartmouth/2018/1/31/collagen-the-do-all-protein-how-it-can-benefit-you

Personal Thoughts & Thank You!